

Experiment 31: Pattern Matching Using Prolog

Aim

To implement **pattern matching using Prolog** and understand how Prolog matches facts with queries and binds variables to matching values.

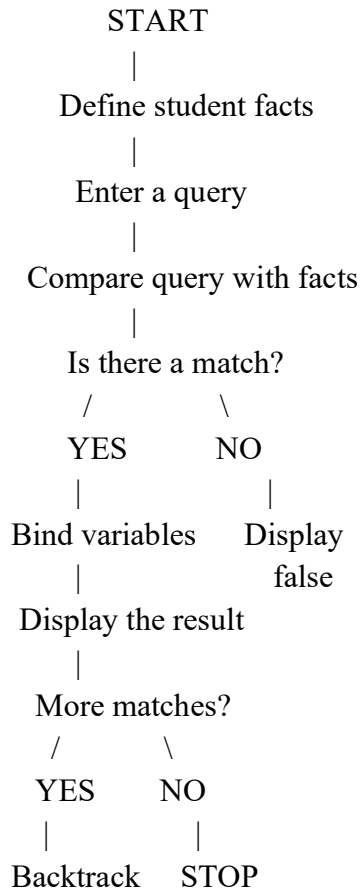
Objective

1. To understand the concept of pattern matching in Prolog.
2. To represent information using Prolog facts.
3. To match specific values with variables.
4. To retrieve information using Prolog queries.
5. To understand how Prolog performs automatic variable binding.

Algorithm

1. Start the program.
2. Define facts representing students and their departments.
3. Define a rule to retrieve student information.
4. Enter a query containing specific values or variables.
5. Prolog compares the query pattern with the available facts.
6. If a matching fact is found, Prolog binds the variables to the corresponding values.
7. Display the matching result.
8. Press ; to find additional matches.
9. Stop when no more matches are available.

Flowchart



Prolog Code

% Pattern Matching in Prolog

```
student(anu, cse).  
student(ravi, ece).  
student(priya, cse).  
student(kumar, it).
```

```
% Rule for pattern matching  
find_student(Name, Department) :-  
    student(Name, Department).
```

Sample Output

Query 1: Match a specific student

```
?- student(anu, cse).
```

Query 2: Find the department of Anu

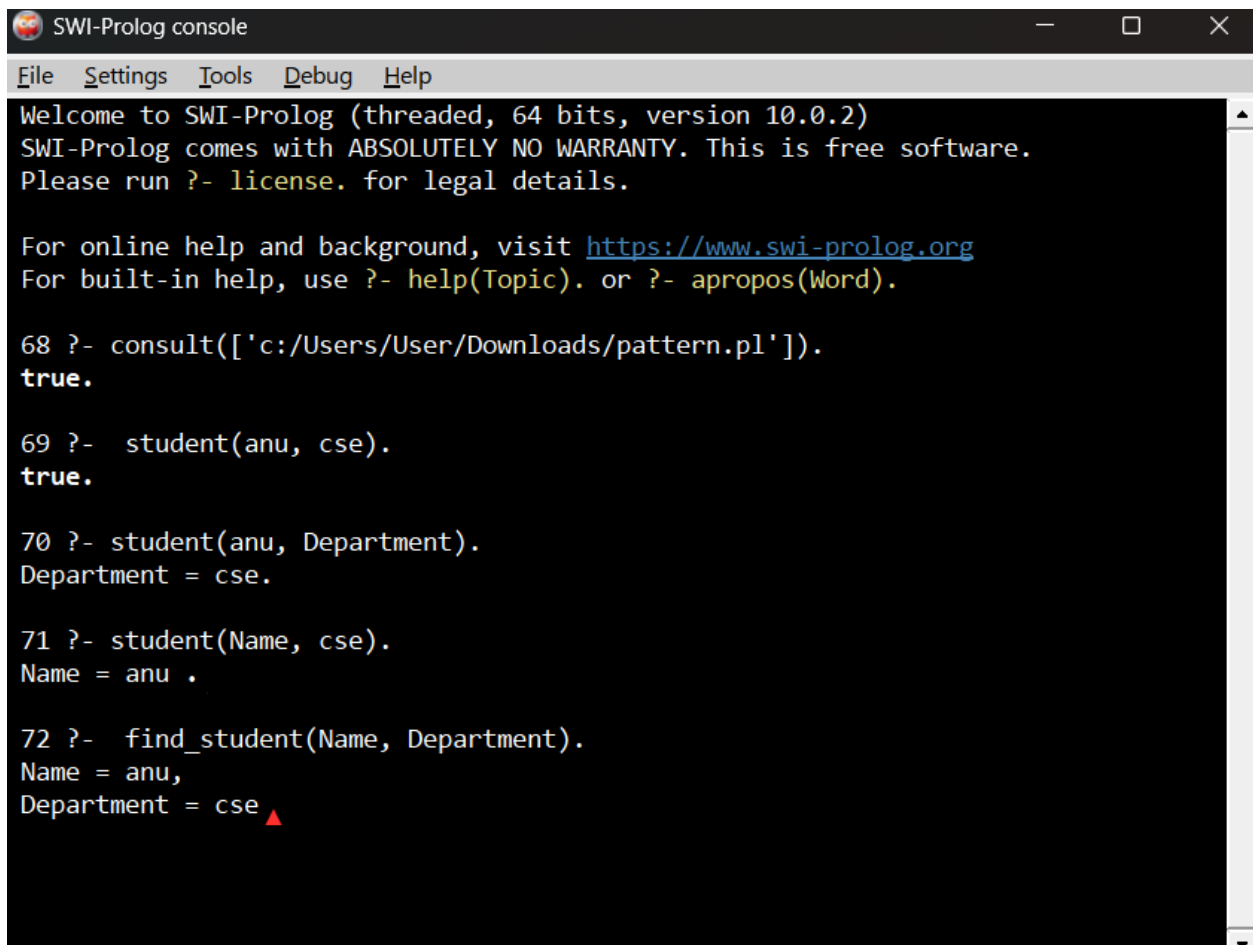
```
?- student(anu, Department).
```

Query 3: Find all CSE students

```
?- student(Name, cse).
```

Query 4: Find all students and their departments

```
?- find_student(Name, Department).
```

A screenshot of the SWI-Prolog console window. The window has a title bar with the SWI-Prolog logo and the text "SWI-Prolog console". Below the title bar is a menu bar with "File", "Settings", "Tools", "Debug", and "Help". The main area of the window is a black terminal with white text. It shows the welcome message for SWI-Prolog (version 10.0.2) and instructions on how to use the console. It then shows the execution of several Prolog queries: 1. A query to consult a file, which returns "true.". 2. A query to find the department of 'anu', which returns "true.". 3. A query to find the department of 'anu', which returns "Department = cse.". 4. A query to find all CSE students, which returns "Name = anu.". 5. A query to find all students and their departments, which returns "Name = anu, Department = cse" followed by a red triangle cursor.

```
SWI-Prolog console
File Settings Tools Debug Help
Welcome to SWI-Prolog (threaded, 64 bits, version 10.0.2)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

68 ?- consult(['c:/Users/User/Downloads/pattern.pl']).
true.

69 ?- student(anu, cse).
true.

70 ?- student(anu, Department).
Department = cse.

71 ?- student(Name, cse).
Name = anu .

72 ?- find_student(Name, Department).
Name = anu,
Department = cse ▲
```

Result

The **Pattern Matching program was successfully implemented using Prolog**. The program correctly matched queries with the available facts and automatically bound variables to the corresponding matching values.

Conclusion

Thus, the experiment demonstrates the use of **pattern matching and variable binding in Prolog**. Prolog compares a query with facts in the knowledge base and returns all matching solutions through logical inference and backtracking.